

Sara Mahmoud

Machine Learning Engineer

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EDUCATION

Pharos University

Faculty of Computer Science and Artificial Intelligence

Alexandria

Oct 2023 – July 2027

GPA

3.84/4

EXPERIENCE

Undergraduate Programming Instructor

Oct 2025 – Present

DesCode Academy

- Taught Python fundamentals and basic programming concepts to pre-university students
- Introduced children to programming using Pictoblox block coding for game development
- Simplified complex technical concepts for beginners and young learners

PROJECTS

Lung Disease Detection and Chatbot | Python, CNN, TensorFlow, Keras, Cv2, NLP, Speech Recognition, Text-to-Speech

- Developed a CNN-based model to detect and classify lung diseases from medical images.
- Built an intelligent chatbot that provides information about detected diseases
- Integrated voice recognition functionality to allow users to interact with the chatbot using spoken questions.
- Implemented a text-to-speech system enabling the assistant to respond with voice output.
- Designed an end-to-end workflow from disease prediction to interactive user support..

Face Recognition System | Python, OpenCV, Face Recognition, NumPy

- Developed a face recognition system capable of identifying individuals from stored facial images.
- Built a dataset of facial images associated with user identities for recognition purposes.
- Implemented real-time face detection and recognition using computer vision techniques.
- Matched detected faces against registered users and displayed the corresponding identity

NFC-Based Smart Attendance System | Python, Selenium, NFC, Web Automation

- Designed and developed an NFC-based attendance management system
- Integrated NFC card UID verification with a university attendance platform.
- Automated student attendance registration using Selenium-based web automation.
- Streamlined the attendance process and minimized manual data entry.

Real-Time Object Detection | Python, YOLOv8, OpenCV, Ultralytics, NumPy

- Built a real-time object detection application using the YOLOv8 deep learning model.
- Detected and classified multiple objects from live video feeds with high accuracy.
- Displayed object labels, confidence scores, and bounding boxes in real time.
- Integrated OpenCV for video capture, processing, and visualization.
- Improved understanding of deep learning and computer vision applications in real-world scenarios.

TECHNICAL SKILLS

Languages: Python, C sharp , HTML/CSS

Machine Learning and AI: Scikit-learn, TensorFlow, Keras, CNN, NLP, Deep Learning, Model Evaluation

Data Analysis and Visualization: Pandas, NumPy, Matplotlib, Seaborn

Developer Tools: Jupyter Notebook, Google Colab, VS Code

Computer Vision: OpenCV, Face Recognition, Image Processing

ACTIVITIES

Head of Media Committee | *HultPrize*

- Managed a team of graphic designers and photographers, ensuring timely delivery of creative media content.

Ex PR member | *HultPrize*

- Assisted in managing public relations activities, communicating with participants and supporting the organization of community events